



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2025

INFORMATION AND COMMUNICATIONS TECHNOLOGY (N540)

Candidates answer on the Question Paper.

TIME: 1 HOUR 45 MINUTES

INSTRUCTIONS TO CANDIDATES

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Any rough working should be done in this booklet.
5. Do NOT use calculators.
6. Answer **ALL** questions.
7. This document consists of **11** questions printed on **24** pages, numbered **2** to **25**.
8. The number of marks is given in brackets [] at the end of each question or part question.
9. The total marks for this paper is **100**.

For Examiners' use													
Question No.	1	2	3	4	5	6	7	8	9	10	11	Total	Signature
Examiner													
Team Leader													
CE/ACE													

SECTION A**Question 1 (15 marks)**

Circle the correct answer. Each item carries one mark.

(a) Which of the following is an **input device**?

A

Projector

B

Monitor

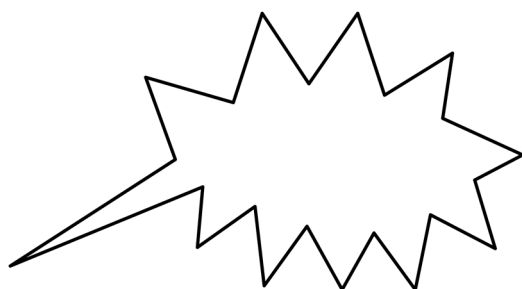
C

Mouse

D

Speaker

(b) Which type of **speech balloon** is shown below?



- A Expression text balloon
- B Whisper balloon
- C Thought balloon
- D General speech balloon

(c) Which one of the following can be used to **view web pages** on the internet?



Microsoft Word



Microsoft Paint



Google Chrome

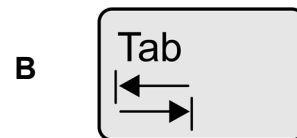
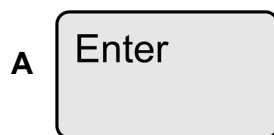


Linux

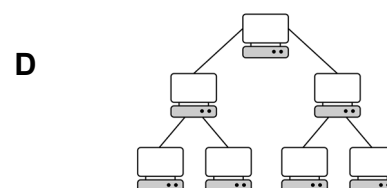
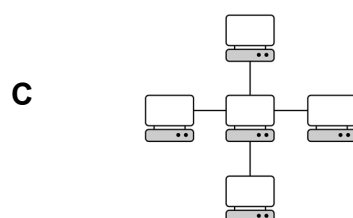
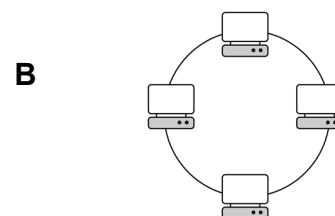
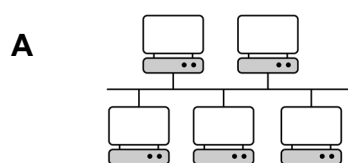
(d) How many **bits** are there in one byte?

- A 4
- B 8
- C 12
- D 16

(e) Which **key** should be pressed to begin a new paragraph in a word processing program?



(f) Which of the following illustrates a **bus topology**?



(g) Which **function** must be used to find the **lowest** value from a range of cells in spreadsheet?

- A MAX
- B AVERAGE
- C MIN
- D SUM

(h) Which **sign** is used to start a **formula** in spreadsheet?

- A +
- B =
- C -
- D /

(i) Which spreadsheet feature is best used to **duplicate information** from a cell?

- A Fill handle
- B Autosum
- C Filter
- D Save as

- (j) In a word processor, what is the **function** of the following icons?



- A** Set the line spacing
- B** Add extra lines
- C** Copy and paste the text
- D** Change alignment
- (k) Which term is used for the plan that contains images and instructions to create a video?
- A** Presentation
- B** Letter
- C** Storyboard
- D** Recorded narration
- (l) How is the **set of principles** that regulates the use of computers in everyday life called?
- A** Computer ethics
- B** Computer crime
- C** Computer piracy
- D** Computer privacy

(m) In a database, which one of the following is a group of records?

- A File
- B Report
- C Field
- D Table

(n) Which of the following is a **search engine**?



(o) What is an **email attachment**?

- A A forwarded email
- B A deleted email
- C A reply to an email
- D A file sent with an email

[15]

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Question 2 (10 marks)

- (a) Fill in the blanks with the **correct term** from the given list.

sound card	motherboard	fan
graphics card	CPU	RAM

- (i) The is the brain of the computer and is responsible for all data processing.
- (ii) The allows us to use microphones with a computer system.
- (iii) The keeps the processor cool and stops it from overheating.
- (iv) The enables the computer to display images.
- (v) The is where all the components of a computer system are connected.

[5]

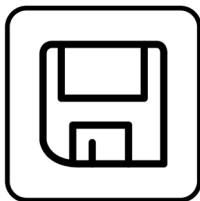
(b) Give the **output on screen** when the following actions are taken.

	Actions	Output on screen
(i)	Pressing Shift + & 7	
(ii)	Pressing (9	

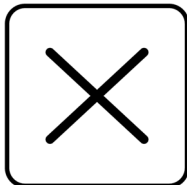
[2]

(c) Match each button in **Column A** to its correct function in **Column B**.

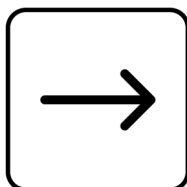
Column A



•



•



•

Column B

•

Close

•

Move cursor to right

•

Move cursor to left

•

Save

[3]

Question 3 (10 marks)

Tick (✓) **True** or **False** next to each statement below. An example is given.

		True	False
(a)	ROM holds programs currently being used.		✓
(b)	A text overlay places an audio file over a video.		
(c)	Hacking means gaining unauthorised access to a computer.		
(d)	A meta-search engine has its own database of websites.		
(e)	A joystick can be used to play games.		
(f)	Megabyte is a smaller unit than Gigabyte.		
(g)	In a comic strip, caption provides information about a panel.		
(h)	During a presentation, the speaker notes are visible to the audience.		
(i)	Spreadsheets can be used to produce pie charts.		
(j)	Header refers to text appearing at the top of the document.		
(k)	Bullet points and numbering are used to create lists.		

[10]

Question 4 (10 marks)

- (a) Match each term in **Column A** to the correct statement in **Column B**.

Write your answers in the **answer grid** provided below.

An example is given.

Column A	
(i)	Cache memory
(ii)	RAM
(iii)	Nibble
(iv)	Bit
(v)	Sensor
(vi)	Plotter
(vii)	Hard copy

Column B	
1	is non-volatile
2	consists of 4 bits
3	can be 0 or 1
4	primary storage
5	is a printout
6	is an input device
7	is volatile
8	is an output device

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
4						

[6]

(b) Label the pictures using the appropriate words from the given list.

Input device	Storage device	Output device	Sprite	Antivirus software
--------------	----------------	---------------	--------	--------------------

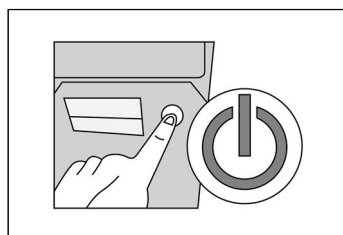
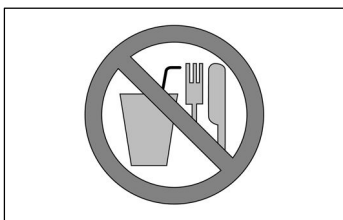
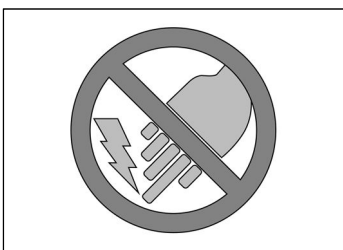
	
	
	
	

[4]

Question 5 (5 marks)

Safety measures that must be adopted in the computer lab are shown below.

Match each picture with the appropriate safety measure.



Do not run in the computer lab

Follow proper steps in switching on or off your computer

Do not touch wires

Adopt a proper sitting posture

Avoid trailing wires in the computer lab

No food or drinks in the computer lab

[5]

SECTION B**Question 6 (5 marks)**

The specifications of a computer system are shown below.

— □ X

System > About

ⓘ Device Specifications

Copy ^

Device name	vm-w11-ssd
Processor	AMD Ryzen Threadripper 2950X-16-Core Processor 3.49 GHz
Installed RAM	16.0 GB
Device ID	89911EB-88F2-4FE0-96EA-98D30CADE640
Product ID	00331-20302-64651-AA251
System type	64-bit operating system x64-based processor
Pen and touch	No pen or touch input is available for this display

Related links

Domain or workgroup

System protection

Advanced system settings

■ ■ Windows specifications

Copy ^

Edition	Windows 11 Pro
Version	21H2
Installed on	10/21/2021
OS Build	22000.829
Experience	Windows Feature Experience Pack 1000.22000.829.0

Microsoft Services Agreement

Microsoft Software License Terms

(a) Give the **name** of the computer.

..... [1]

(b) Tick (✓) the correct **model** of the processor used in the laptop.

Celeron ☐ AMD ☐ Intel ☐ [1]

(c) What is the **speed** of the processor used in the laptop?

..... [1]

(d) Give the name of the **operating system** installed on the computer.

..... [1]

(e) What is the **memory size** of the computer system?

..... [1]

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Question 7 (7 marks)

An extract of a document is given below. Answer the questions that follow.



(a) Which font type has been used for the title '2016 Summer Olympics'?

..... [1]

(b) What is the font size of the title '2016 Summer Olympics'?

..... [1]

(c) Which alignment has been used for the 2nd paragraph?

..... [1]

(d) Which type of formatting has been used for "Kosovo" in **line 2** of the second paragraph?

Underline ☐

Italic ☐

Bold ☐

[1]

- (e) A hard copy of the document must be printed.
The diagram below shows the settings of the printer.

Print

Copies: 5

Printer

hp deskjet 3600 series (LIDIL)
Ready

[Printer Properties](#)

Settings

Print Custom Range
Enter specific pages or section...

Pages: 4 - 10

Print One Sided
Only print on one side of the p...

Collated
1,2,3 1,2,3 1,2,3

Portrait Orientation

A5
14.8 cm x 21 cm

Custom Margins

State

- (i) the number of copies that will be printed,
..... [1]
- (ii) the size of the paper selected,
..... [1]
- (iii) the pages of the document that will be printed.
..... [1]

Question 8 (9 marks)

Jane is comparing prices of PC hardware from three different shops and is using a spreadsheet to calculate the differences, as illustrated in the image below.

	A	B	C	D	E	F	G	H
1	Price comparison of PC hardware							
2								
3	Items	Shop A	Shop B	Shop C	Highest	Lowest	Average	
4	Pendrive	€ 10.00	€ 12.50	€ 16.00	€ 16.00	€ 10.00	€ 12.83	
5	Mouse	€ 12.50	€ 15.00	€ 13.00	€ 15.00	€ 12.50	€ 13.50	
6	Keyboard	€ 25.00	€ 23.00	€ 19.00	€ 25.00	€ 19.00	€ 22.33	
7	Headset	€ 75.00	€ 85.00	€ 96.00	€ 96.00	€ 75.00	€ 85.33	
8								
9	Total Excluding VAT	€122.50	€135.50	€144.00				
10	VAT at 18%	€ 22.05	€ 24.39	€ 25.92				
11								
12	Total Including VAT	€144.55	€159.89	€169.92				
13								

- (a) Write the content of cell **A7**.

..... [1]

- (b) Give the **cell address** that contains the price of a mouse in Shop B.

..... [1]

- (c) Using an appropriate function, write a formula that should be entered in cell **B9** to find the **total excluding VAT** for the items purchased in Shop A.

.....[2]

- (d) Tick (✓) the **correct formula** that has been used in cell **B10** to calculate the VAT at 18% on the items purchased from Shop A.

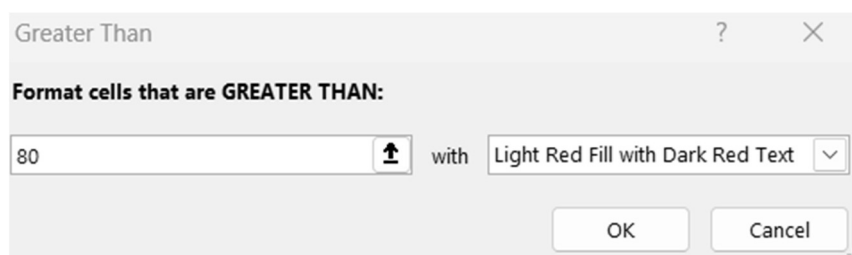
= B9 + 0.18 ☐
 = B9 x 0.18 ☐
 = B9 * 0.18 ☐

[1]

- (e) In cell **H4**, a formula was entered to find the **average price** of a pendrive. Write down the formula.

..... [2]

- (f) Conditional formatting is applied to cell **B4:D7** as follows:



Name the **cells** that will be formatted.

..... [2]

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Question 9 (9 marks)

Video conferencing is becoming very popular nowadays.

- (a) Explain what is meant by **video conferencing**.

.....

.....

..... [2]

- (b) Name **two** devices apart from **keyboard**, **mouse** and **monitor** that are required during video conferencing.

1.

2. [2]

- (c) State **two** advantages of using video conferencing.

1.

2. [2]

- (d) State **one** disadvantage of using video conferencing.

..... [1]

- (e) State **two** uses of video conferencing.

1.

2. [2]

E
TL
CE

Question 10 (10 marks)

- (a) (i) Give **two** advantages of using a computerised database program.

.....
..... [2]

- (ii) A list of sports is given below:

- Badminton
- Football
- Tennis
- Cricket

If the data given above is sorted in **descending order**, which sport would be first in the list?

..... [1]

- (b) A hairdresser uses a database to store data on upcoming appointments of clients. Part of the database is shown below.

The **APPOINTMENT** table is shown below.

APT_ID	DATE	TIME	NAME	STYLE
JP15	1/2/2026	09:45	Saleem	Cut
JP12	1/2/2026	14:00	Vicky	Cut and Dry
JP13	1/2/2026	14:30	Vidya	Colour
AD14	1/2/2026	16:00	Karen	Cut and Colour
JP16	2/2/2026	11:15	Steven	Cut and Dry
AD17	2/2/2026	12:30	Daniela	Colour

- (i) The following search condition was entered using a **query by example** grid.

Field	APT_ID	STYLE
Table	APPOINTMENT	APPOINTMENT
Sort	ASCENDING	
Show	☒	☐
Criteria		= "Colour"
Or		

Show what the **output** would be.

.....

.....

..... [3]

- (ii) Complete the **query by example** grid below to select and show the names of clients who have appointments on 2/2/2026.

Field		
Table	APPOINTMENT	APPOINTMENT
Sort		
Show	☐	☐
Criteria		
Or		

[4]

E

TL

CE

Question 11 (10 marks)

- (a) Fill in the blanks using the terms given below.

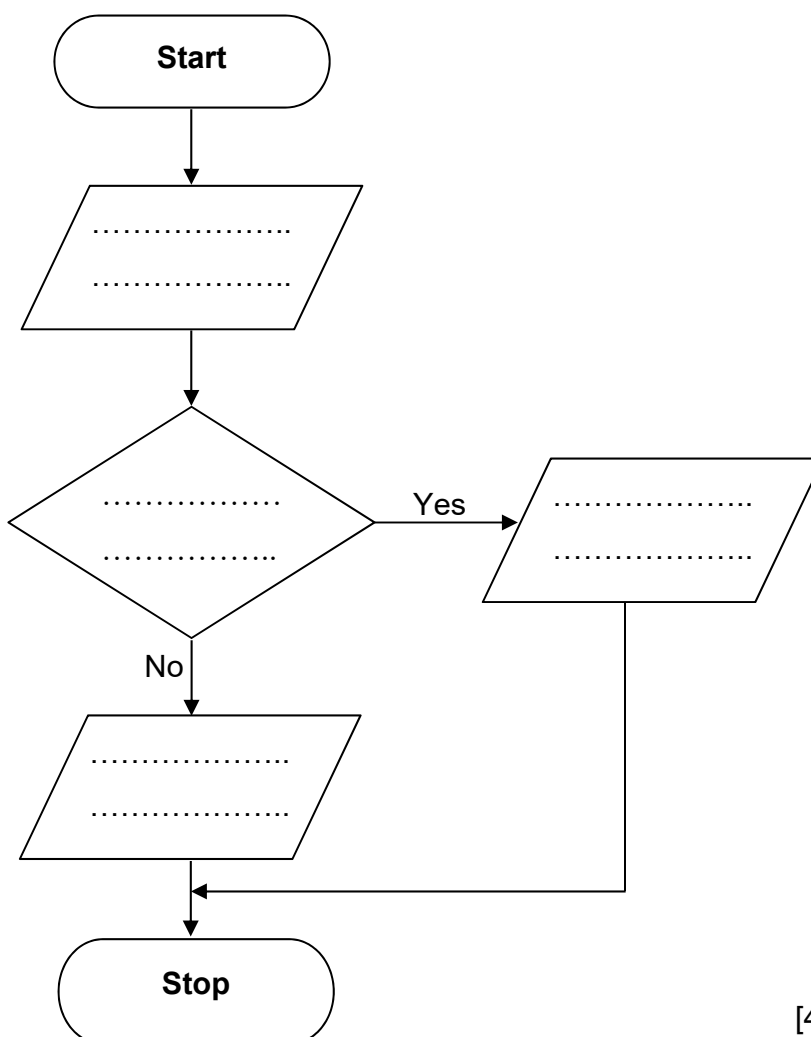
rectangle	diamond
circle	parallelogram

In a flowchart,

- (i) a is used to represent input and output.
 (ii) a is used to represent a process.
 (iii) a is used to represent a decision box.

[3]

- (b) The flowchart below allows the user to enter a mark for an ICT test.
 If the mark is **35 or more**, then output **Pass**, else output **Fail**.
 Complete the flowchart.



[4]

- (c) Write statements in Python that will allow a user to enter a value in the variable marks and then output its contents.

.....

.....

..... [3]

END OF PAPER

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